

TERNOPIL, Ukraine

WMO: 334150

Lat: 49.5278N

Lon: 25.6911E

Elev: 329

StdP: 97.43

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.9	-15.3	-21.3	0.6	-18.6	-17.9	0.8	-15.0	11.9	0.2	10.3	-0.5	1.8	20	0.595

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	29.6	19.8	27.9	19.0	26.3	18.4	21.0	27.4	20.1	26.0	19.2	24.6	4.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.8	14.2	23.7	18.0	13.4	22.8	17.2	12.8	21.9	62.2	27.5	59.0	26.0	56.1	24.8	24.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.8	8.9	8.2	-22.0	32.1	4.3	1.5	-25.1	33.2	-27.6	34.0	-30.0	34.8	-33.1	35.9
			-22.2	22.4	4.3	0.8	-25.2	23.0	-27.7	23.5	-30.1	24.0	-33.2	24.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.0	-4.1	-2.5	1.9	8.9	14.2	17.2	19.2	18.6	13.7	8.1	2.8	-2.4
	DBStd	9.35	5.68	5.83	4.87	4.23	3.70	3.19	2.95	3.04	3.61	4.18	4.55	5.29
	HDD10.0	1810	436	351	251	68	8	0	0	0	8	87	218	383
	HDD18.3	3892	694	584	508	283	136	58	26	33	143	318	466	642
	CDD10.0	1092	0	0	1	35	137	217	284	267	120	28	3	0
	CDD18.3	133	0	0	0	1	7	26	52	43	6	0	0	0
	CDH23.3	1243	0	0	0	8	84	207	456	418	70	1	0	0
	CDH26.7	288	0	0	0	1	8	28	118	121	12	0	0	0
(14) Wind	WSAvg	3.9	4.3	4.4	4.5	4.2	3.9	3.7	3.4	3.2	3.6	3.9	4.2	4.2
(15) Precipitation	PrecAvg	610	33	33	28	44	69	79	85	64	55	34	37	38
	PrecMax	815	107	82	69	115	122	183	209	181	181	99	148	67
	PrecMin	396	5	2	6	7	17	19	11	16	2	4	0	6
	PrecStd	102	25	21	19	27	29	38	40	31	40	24	26	16
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.9	11.3	16.6	24.4	27.9	29.2	31.9	32.0	28.2	22.4	15.8	8.9
		MCWB	5.5	7.3	9.9	14.7	17.8	19.9	21.1	20.1	17.9	15.2	11.5	6.8
	2%	DB	5.5	7.7	13.1	21.0	25.7	27.4	29.8	29.8	25.3	19.3	12.6	7.0
		MCWB	3.6	5.5	8.2	13.1	16.8	19.0	20.2	19.6	16.5	13.6	10.2	5.7
	5%	DB	3.9	5.7	10.5	18.6	23.7	25.9	27.8	27.7	22.9	16.5	10.7	5.1
		MCWB	2.5	4.0	6.6	11.8	15.9	18.3	19.5	19.0	15.5	12.4	9.0	4.0
	10%	DB	2.5	3.9	8.4	16.2	21.6	24.1	25.9	25.6	20.5	14.3	9.1	3.3
		MCWB	1.4	2.5	5.7	10.4	15.0	17.8	18.8	18.3	14.8	11.4	7.7	2.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.8	7.8	10.3	15.3	19.1	21.4	22.4	22.1	18.8	16.2	11.9	7.6
		MCDB	7.5	11.2	15.5	22.7	25.6	27.1	29.0	28.7	26.4	20.9	14.6	8.7
	2%	WB	3.9	5.6	8.5	13.8	17.7	20.2	21.2	20.8	17.4	14.2	10.6	5.6
		MCDB	5.1	7.5	12.4	20.0	23.7	25.5	27.7	27.5	22.9	18.1	12.4	6.8
	5%	WB	2.7	4.1	7.1	12.3	16.6	19.2	20.2	19.7	16.3	12.9	9.2	4.0
		MCDB	3.7	5.6	9.9	17.3	22.3	24.2	26.1	25.7	21.2	15.9	10.6	5.0
	10%	WB	1.5	2.6	5.8	11.1	15.5	18.3	19.4	18.8	15.2	11.5	7.7	2.6
		MCDB	2.4	3.8	8.2	15.4	20.5	22.9	24.7	24.1	19.7	14.0	8.9	3.3
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	5.4	6.8	9.4	10.1	9.7	10.1	10.6	9.4	7.9	5.1	4.6
		MCDBR	4.8	7.2	11.2	13.4	13.7	12.9	13.6	14.5	14.1	12.2	8.5	5.9
		MCWBR	3.8	5.3	7.0	7.0	6.4	5.9	5.6	5.9	6.5	7.0	5.9	4.8
	5% WB	MCDBR	4.4	7.0	9.9	12.1	12.2	11.0	11.9	12.6	12.2	10.7	7.8	5.6
		MCWBR	3.7	5.3	6.6	6.6	6.0	5.5	5.3	5.7	6.2	6.5	5.7	4.6
(40) Clear-Sky Solar Irradiance	taub		0.309	0.339	0.379	0.430	0.412	0.411	0.439	0.439	0.399	0.364	0.340	0.306
	taud		2.238	2.195	2.192	2.126	2.247	2.269	2.201	2.199	2.269	2.350	2.339	2.309
	Ebn at Noon		719	785	817	812	848	850	817	796	789	755	681	673
	Edh at Noon		74	100	119	142	131	129	136	131	110	86	69	61
(44) All-Sky Solar Radiation	RadAvg		0.93	1.75	2.88	4.22	5.31	5.66	5.47	4.89	3.40	2.10	0.95	0.66
	RadStd		0.14	0.25	0.38	0.44	0.52	0.41	0.41	0.40	0.52	0.31	0.11	0.11

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.65	N/A	N/A	+0.85	+0.56	N/A	N/A	-208	+120	+46
(47) Regional (10 neighbors)		+0.65	N/A	N/A	+0.95	+0.46	N/A	N/A	-200	+118	+48

Nomenclature: See separate page